Phase-Space Volume Contraction, Not the 0–1 Test, Separates Conservative from Dissipative Chaos: A Cross-System Cautionary Result

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Abstract

Deterministic chaos defeats long-horizon trajectory prediction, but many structural features of a chaotic system remain measurable. We test whether a cheap diagnostic can distinguish conservative Hamiltonian chaos from dissipative chaos, rather than merely detecting chaos itself. Across a 26-system zoo (chaotic maps, dissipative flows, Hamiltonian flows, a delay system, and non-chaotic controls), an earlier version of this study appeared to show that the Gottwald–Melbourne 0–1 statistic separates conservative from dissipative systems. A trajectory-length audit falsified that conclusion: for chaotic Hamiltonian systems the statistic is length-dependent and can climb from near zero to near one (double pendulum: $K = -0.18$ at 4k steps to $K = 0.98$ at 20k steps). The 0–1 statistic is therefore a chaos detector, not a conservative/dissipative classifier. The robust discriminator is instead the phase-space volume-contraction rate $\Sigma\lambda = \langle \nabla \cdot f \rangle$, equal to the sum of Lyapunov exponents: it is zero to within 10^{-3} for all 8 Hamiltonian systems and strictly negative for all 11 finite-dimensional dissipative flows with defined vector fields, matching analytic traces where available. We further show why the false 0–1 split is seductive: the entire K-vs-length curve is predicted from the observable autocovariance by a closed-form Green–Kubo expression (Pearson $r = 0.955$ against measured K). Finally, a held-out symbolic-itinerary experiment shows that attractor transition grammar is predictable (0.770 vs. 0.233 baseline on cell changes, 21/23 chaotic systems beating baseline) even when microstate forecasts are Lyapunov-limited. The contribution is a corrected discriminator, a reproducible warning about finite-time misuse of a widely used chaos statistic, and a bounded constructive result about "predicting the climate, not the weather."

1 Introduction

Chaotic systems share sensitivity to initial conditions, yet they do not share the same geometric or physical structure. A Hamiltonian double pendulum preserves phase-space volume and remains on an energy shell; the Lorenz system contracts onto a strange attractor. A scalar chaos detector can report that both systems are chaotic without answering whether the chaos is conservative or dissipative.

This paper asks a narrow question: among low-cost diagnostics, what separates conservative Hamiltonian chaos from dissipative chaos? We tested two candidates. The first was the Gottwald–Melbourne 0–1 test, which maps a scalar time series to a statistic K approaching one for chaotic dynamics and zero for regular dynamics [1, 2]. The second was phase-space volume contraction, $\Sigma\lambda = \langle \nabla \cdot f \rangle$, the divergence of the vector field averaged along the trajectory, equal to the sum of Lyapunov exponents for smooth flows [3, 4]. Liouville's theorem makes the latter a natural discriminator: Hamiltonian flows preserve phase-space volume, while dissipative flows contract it.

The important part is that one candidate initially fooled us. Early runs at short trajectory lengths suggested that conservative chaos had $K \approx 0$ while dissipative chaos had $K \approx 1$. Longer trajectories and initial-condition ensembles overturned that reading. We therefore present the correction as part of the result: the 0–1 test remains useful as a chaos detector, but it should not be used as a conservative/dissipative classifier without additional structure.

2 Systems and measurements

The benchmark contains 26 systems spanning maps, dissipative flows, Hamiltonian flows, delay dynamics, and non-chaotic controls (Table 1). Flows are integrated by fixed-step RK4 after warm-up onto the attractor; maps are iterated directly. Mechanical Hamiltonian systems are integrated in canonical (q, p) coordinates. Known analytic invariants are tracked as integration hygiene checks.

0–1 test. For a scalar observable $\phi(j)$ and frequency c, define translation variables

$$p_c(n) = \sum_{j=1}^{n} \phi(j) \cos(jc), \qquad q_c(n) = \sum_{j=1}^{n} \phi(j) \sin(jc). \tag{1}$$

The mean-square displacement $M_c(n)$ is computed over lags, and $K_c = \mathrm{corr}(\log n, \log M_c(n))$ is averaged over random c. In the asymptotic regime, $K \to 1$ for chaotic dynamics and $K \to 0$ for regular dynamics. The finite-length regime is the failure mode studied here.

Phase-space volume contraction. For a vector field $\dot{x} = f(x)$,

$$\Sigma\lambda = \langle \nabla \cdot f(x(t)) \rangle = \left\langle \sum_i \frac{\partial f_i}{\partial x_i} \right\rangle. \tag{2}$$

For smooth flows this equals the sum of Lyapunov exponents. Hamiltonian flows have $\nabla \cdot f = 0$ in canonical coordinates; dissipative flows contract volume. Where analytic traces are available, they are used as cross-checks against the numerical central-difference estimator.

3 Results

3.1 The 0–1 statistic is not a conservative/dissipative discriminator

On sufficiently long trajectories, Hamiltonian K values scatter across the same range as dissipative chaos. The decisive evidence is the length audit (Fig. 1). Dissipative systems such as Lorenz and Chen read $K \approx 1$ at all tested lengths. Hamiltonian systems can begin near zero and climb toward one as the trajectory length grows. The double pendulum moves from $K \approx -0.13$ (mean over 4 initial conditions) at 4k steps to $K \approx 0.99$ at 16k steps; the coupled Duffing Hamiltonian moves from $K \approx -0.14$ at 4k to $K \approx 0.94$ at 16k. This is the pattern one expects from a chaos detector whose finite-time behavior is governed by observable mixing time, not by whether the system is conservative.

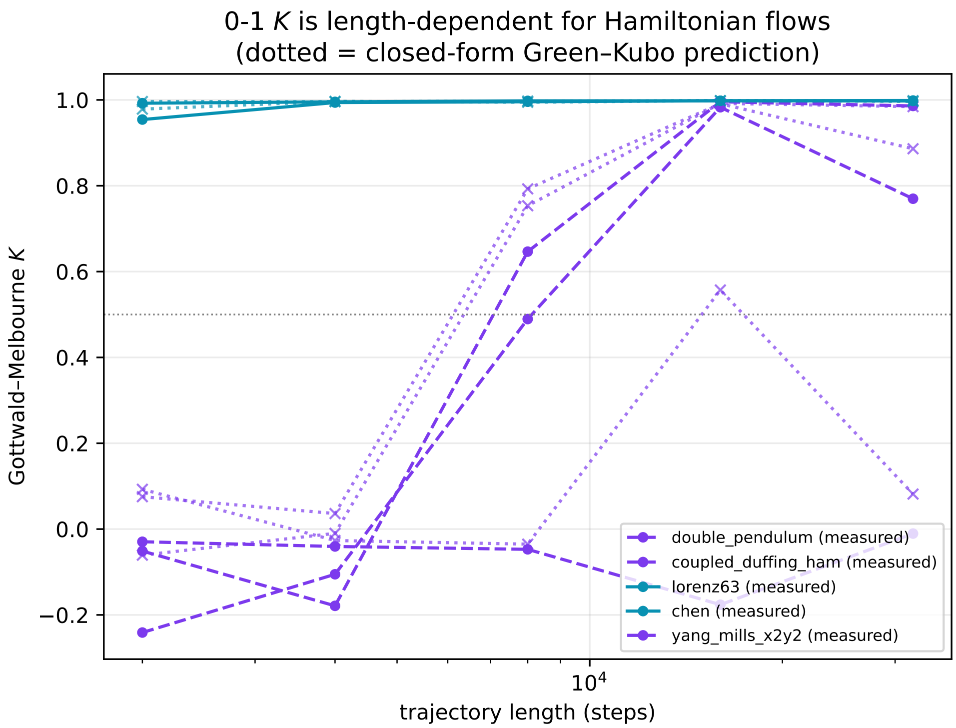

Figure 1: The 0–1 statistic is trajectory-length dependent for slow-mixing Hamiltonian systems. Purple curves can climb from near zero to near one; cyan dissipative systems read near one at all lengths. Dotted curves are the closed-form Green–Kubo predictions.

3.2 Volume contraction separates the classes

The phase-space contraction rate cleanly separates the finite-dimensional systems with defined vector fields (Fig. 2). The three maps and the Mackey–Glass delay system are excluded from this analysis because they lack a finite-dimensional smooth vector field; $\Sigma\lambda$ is defined only for ordinary differential equation systems. Among those, all 8 Hamiltonian flows have $\Sigma\lambda = 0$ to within 10^{-3}, and all 11 dissipative flows have $\Sigma\lambda < 0$. The estimator reproduces analytic traces where they exist: Lorenz63 gives $-13.667 = -\sigma - 1 - \beta$, Chen gives -10, Lorenz96$_N$ gives $-N$, Sprott B gives -1, Thomas gives $-3b$, and Ueda gives $-k$. Because $\Sigma\lambda$ is a vector-field property rather than a finite scalar-series random walk, it has no trajectory-length crossover analogous to the 0–1 statistic.

The Hamiltonian cases also pass integration hygiene: analytic invariant drift is at most 7×10^{-4} in the generated artifacts, and the measured largest Lyapunov exponent is positive for every chaotic Hamiltonian system. Thus the zero volume contraction is not a failure to reach chaos; it is the expected Liouville structure.

3.3 Why the 0–1 failure happens

The finite-time behavior of K is not mysterious. For a mean-zero stationary observable with autocovariance $C(k) = E[\phi(j)\phi(j+k)]$, the 0–1 translation increment has exact mean-square displacement

$$M_c(n) = \sum_{k=-(n-1)}^{n-1} (n - |k|)C(k)\cos(kc) = nC(0) + 2\sum_{k=1}^{n-1}(n-k)C(k)\cos(kc). \tag{3}$$

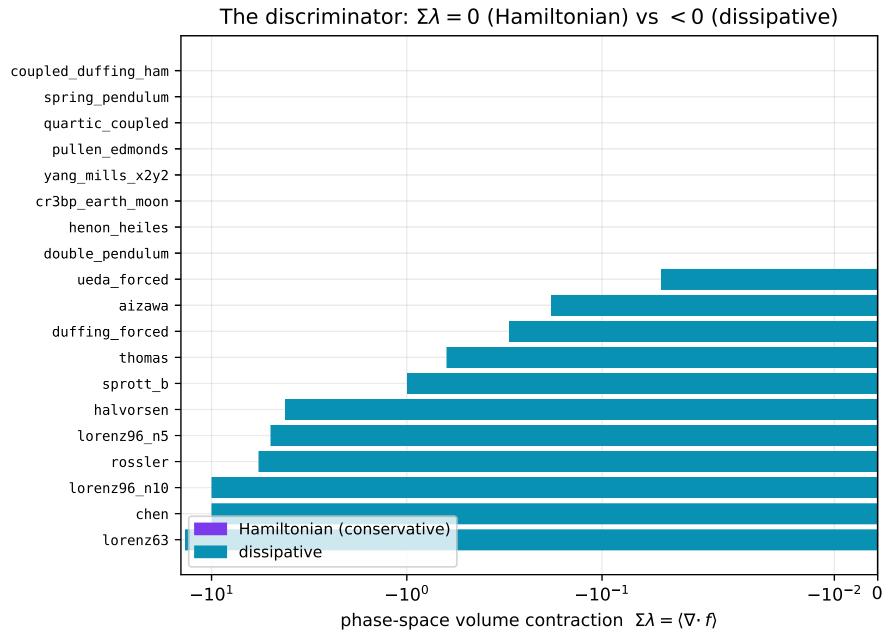

Figure 2: Phase-space volume contraction separates conservative from dissipative chaos. Hamiltonian systems sit at zero; dissipative flows contract. The horizontal axis is symlog so both weak and strong contraction are visible.

Its asymptotic diffusion coefficient is the Green–Kubo / Wiener–Khinchin power spectral density at frequency c [5, 6]:

$$D_c = \lim_{n\to\infty} \frac{M_c(n)}{n} = C(0) + 2\sum_{k\geq 1} C(k)\cos(kc). \tag{4}$$

Since K_c is the log–log slope/correlation computed from $M_c(n)$, the observable autocovariance predicts the full length-dependent curve. In the artifacts, the predicted K from Eq. (3) matches measured K with Pearson $r = 0.955$ and mean absolute error 0.077 (Fig. 3); the predicted crossover length is within one ladder step on 20/23 chaotic systems. The apparent class split was therefore a mixing-time artifact.

3.4 Held-out itinerary grammar: predict the climate, not the weather

The same observatory also measures symbolic attractor grammar. Each attractor is coarsened into cells by fitting k-means ($k \in \{4, 6, 8\}$ cells tested; results reported at the representative $k = 6$ unless otherwise noted) on the first half of the trajectory; an order-1 Markov model is then evaluated on a held-out second half. Raw next-symbol accuracy across 23 chaotic systems is 0.925, but most of that is trivial dwell-time persistence: finely sampled flows tend to remain in the same cell, and the persistence baseline is 0.863. The non-trivial result conditions on the roughly 14% of steps where the cell changes. On those transitions, the learned grammar predicts the destination cell with mean accuracy 0.770 versus a 0.233 baseline, beating baseline on 21/23 systems (Fig. 4). This is not microstate prediction. It is a bounded, held-out statement that chaotic attractors can have learnable symbolic route grammar.

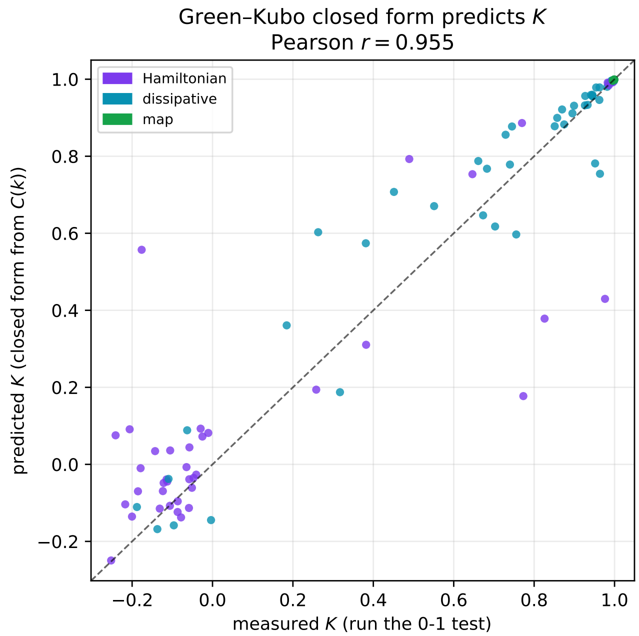

Figure 3: Closed-form Green–Kubo prediction of measured 0–1 K from the observable autocovariance alone.

4 Parameter sweeps and stress checks

Parameter sweeps stress-test the two conclusions (Fig. 5). For Lorenz63, $\nabla \cdot f = -\sigma - 1 - \beta$ is independent of ρ; the measured $\Sigma\lambda$ remains pinned at -13.667 across $\rho \in [28, 150]$ while the largest Lyapunov exponent varies by a factor of about 30. Thus volume contraction measures dissipative structure, not chaos strength. The 0–1 test, by contrast, correctly flags non-chaotic regimes within a family: a Lorenz periodic window and a standard-map KAM regime produce low K. This supports the corrected interpretation: 0–1 is a chaos detector whose finite-time behavior depends on mixing, while $\Sigma\lambda$ is the conservative/dissipative discriminator when the vector field is available.

5 Discussion

The practical message is simple. If one only has a scalar time series and wants to ask whether the dynamics are chaotic, the 0–1 test remains useful. If one has a model vector field and wants to distinguish conservative from dissipative chaos, volume contraction is the theory-grounded diagnostic. The failure mode arises when a finite-time scalar chaos detector is quietly promoted into a structural classifier. Slow mixing then masquerades as conservative structure.

The itinerary result points in a different direction. It does not overcome the Lyapunov wall for trajectories; it changes the prediction target. Local microstate forecasts fail at long horizons, but coarse symbolic route grammar can remain predictable on held-out data. That distinction is useful for scientific machine learning systems because it separates impossible weather-like prediction from bounded climate-like structure.

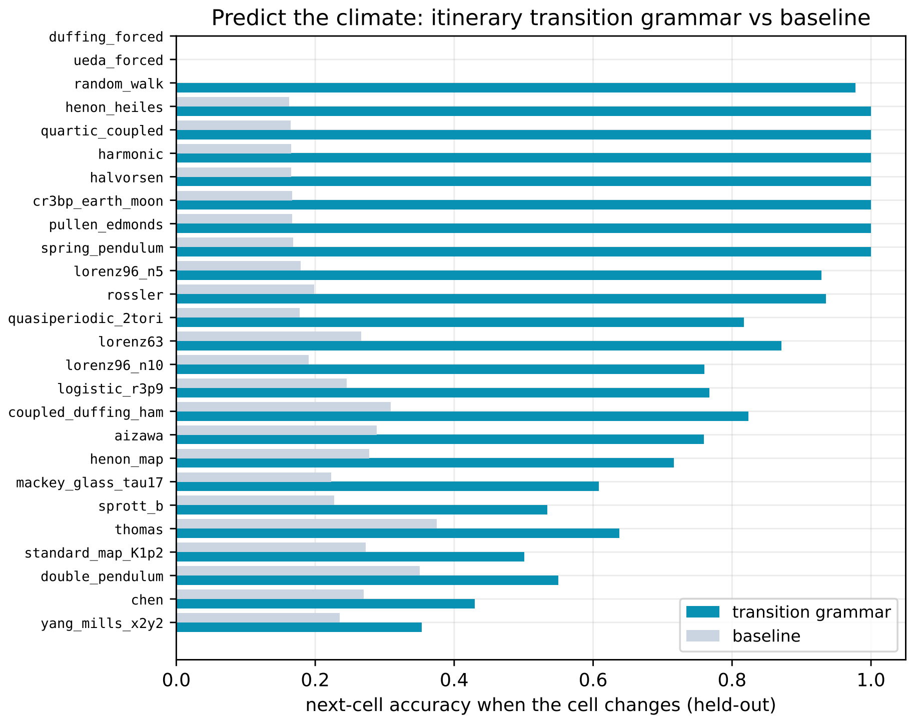

Figure 4: Held-out next-cell prediction when the cell changes. The transition grammar beats the most-common-transition baseline on 21/23 chaotic systems.

6 Limitations and claim boundary

Supported claims are limited to the generated system zoo and the deterministic artifacts listed below. The positive discriminator claim is empirical over 8 Hamiltonian systems and 11 dissipative finite-dimensional flows with defined vector fields; Liouville's theorem supplies the theoretical reason for Hamiltonian zero contraction, but this paper does not prove a new theorem. The cautionary 0–1 claim is stronger in practice: the observed length dependence and Green–Kubo prediction show why the statistic should not be used as a conservative/dissipative classifier. The itinerary result is held-out symbolic transition prediction, not microstate forecasting and not a Foundation-model prediction claim.

Retracted: an earlier preliminary analysis of this dataset incorrectly concluded that conservative chaos has $K \approx 0$; that reading is superseded here. Not supported: using the 0–1 statistic as a conservative/dissipative classifier; generalizing the finite zoo to all chaotic systems without further validation; claiming a new theorem; or claiming long-horizon chaotic trajectory prediction.

7 Reproducibility and data availability

All reported numbers are read from deterministic, seeded artifacts. Data and figure-generation scripts are available at `https://github.com/MonongahelaHellbender/Autonomous-Scientist-Core` (branch `foundation-stable-closeout`). The primary evidence files are:

`results/chaos_universality_lab_best.json`;
`results/chaos_mixing_time_analysis.json`; `results/chaos_green_kubo.json`;
`results/chaos_parameter_sweep.json`; and `results/chaos_itinerary_prediction.json`.

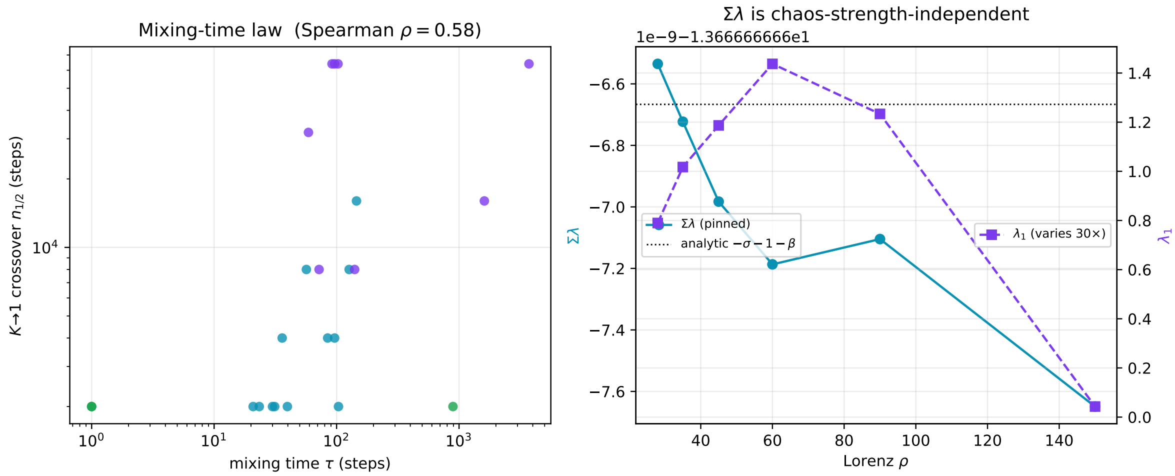

Figure 5: Left: mixing-time law, with longer observable autocorrelation associated with later K crossover. Right: Lorenz volume contraction remains pinned to the analytic trace while λ_1 changes substantially across the parameter sweep.

Figures are regenerated by `scripts/liquid/chaos_make_figures.py`. The manuscript intentionally reports the corrected claim boundary and preserves the self-disproved 0–1 result rather than deleting it.

A Effective-sample interpretation of the 0–1 crossover

Equation (3) implies that K reaches its asymptotic chaotic value only when the trajectory is long relative to the observable's integrated autocorrelation time. The artifacts estimate this time by a Sokal-style windowed autocorrelation procedure [7]. Across the cross-system ladder, Spearman $\rho(\tau, n_{1/2}) = 0.58$: longer observable mixing time delays the first length where $K > 0.5$. Dissipative systems have median crossover near 2k steps, while Hamiltonian systems have median crossover near 16k steps. The relevant variable is therefore effective sample size after decorrelation, not conservation versus dissipation.

B Reference table for the main claim

Question	Correct diagnostic	Boundary
Is the scalar series chaotic?	0–1 K	Requires sufficient effective length
Is the flow conservative or dissipative?	$\Sigma\lambda = \langle \nabla \cdot f \rangle$	Requires vector field/model
Can the exact future state be predicted?	No long-horizon claim	Lyapunov wall remains
Can coarse route grammar be predicted?	Held-out itinerary model	Symbolic transitions only

References

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Table 1: The 26-system benchmark. λ_1: largest Lyapunov exponent (positive = chaotic). K: 0–1 statistic at the full run length. $\Sigma\lambda$: measured phase-space volume contraction (N/A for maps and the delay system, which lack a finite-dimensional smooth vector field).

System	Type	Dim	λ_1	K	$\Sigma\lambda$
Dissipative flows					
Lorenz 63	Dissipative	3	+0.761	0.996	−13.667
Rössler	Dissipative	3	+0.057	0.801	−5.737
Duffing (forced)	Dissipative	3	+0.114	0.729	−0.300
Lorenz 96 ($N = 5$)	Dissipative	5	+0.587	0.944	−5.000
Aizawa	Dissipative	3	+0.122	0.927	−0.183
Sprott B	Dissipative	3	+0.217	0.999	−1.000
Halvorsen	Dissipative	3	+0.856	0.954	−4.200
Thomas	Dissipative	3	+0.046	0.995	−0.625
Chen	Dissipative	3	+2.134	0.998	−10.000
Ueda (forced)	Dissipative	3	+0.100	0.513	−0.050
Lorenz 96 ($N = 10$)	Dissipative	10	+1.129	0.996	−10.000
Hamiltonian flows					
Double pendulum	Hamiltonian	4	+0.888	0.770	0.000
Hénon–Heiles	Hamiltonian	4	+0.026	0.877	0.000
CR3BP (Earth–Moon)	Hamiltonian	4	+0.168	−0.046	0.000
Yang–Mills (x^2y^2)	Hamiltonian	4	+0.241	−0.110	0.000
Pullen–Edmonds	Hamiltonian	4	+0.317	−0.100	0.000
Quartic coupled	Hamiltonian	4	+0.250	0.242	0.000
Spring pendulum	Hamiltonian	4	+0.116	0.979	0.000
Coupled Duffing (Ham.)	Hamiltonian	4	+0.214	0.957	0.000
Maps (no vector field; $\Sigma\lambda$ not applicable)					
Logistic ($r = 3.9$)	Map	1	+1.322	1.000	N/A
Hénon map	Map	2	+0.584	0.999	N/A
Standard map ($K = 1.2$)	Map	2	+1.652	0.996	N/A
Delay system ($\Sigma\lambda$ not applicable)					
Mackey–Glass ($\tau = 17$)	Delay	∞	—	0.855	N/A
Non-chaotic controls					
Harmonic oscillator	Control	2	0.000	0.599	N/A
Quasiperiodic 2-torus	Control	2	0.000	0.961	N/A
Random walk	Control	2	—	0.045	N/A